

FILE HANDLING

File:

A file is a collection of data stored in hard disk.

File Path:

A file path can be either relative or absolute.

An absolute path always contains the root and the complete directory list to specify the exact location of the file.

A relative path starts with respect to the current working directory.

Types of Files:

1) Text Files

2) Binary Files

A text file is a collection of characters.

A binary file is a collection of bytes.

The open() function:

It creates file object and returns. It is used to open a file with specified mode.

Syntax:

```
objectReference=open("FileName", "AccessMode")
```

Example:

```
f=open("abc.txt", "w")
```

List of Access Modes:

r => It is a default mode. It opens a file for reading only in text format.

rb => It opens a file for reading only in binary format.

r+ => It opens a file for both reading & writing in text format.

rb+ => It opens a file for both reading & writing in binary format.

w => It opens a file for writing only in text format.

wb => It opens a file for writing only in binary format.

w+ => It opens a file for both writing & reading in text format.

wb+ => It opens a file for both writing & reading in binary format.

a => It opens a file for appending in text format.

ab => It opens a file for appending in binary format.

a+ => It opens a file for both appending & reading in text format.

ab+ => It opens a file for both appending & reading in binary format.

The close() function:

It is used to close the file associated with the file object.

Syntax:

objectReference.close()

Example:

f.close()

File Methods:

read() => Returns file content

readline() => Returns one line from the file

readlines() => Returns all lines from the file

write() => Writes the specified string to a file

writelines() => Writes the list of strings to a file.

tell() => Returns the file current position

seek() => Moves the file pointer to a specified position

Example1:**#Program to write data to a file**

```
f=open("abc.txt", "w")  
f.write("Welcome To Python")  
f.close()
```

Example2:**#Program to read data from file**

```
f=open("abc.txt", "r")  
x=f.read()  
print(x)  
f.close()
```

Example3:**#Program to append the data to a file**

```
f=open("abc.txt", "a")  
f.write("Python By Venkatesh Mansani")  
f.close()
```

Example4:**#Program to demonstrate tell() & seek() methods**

```
f=open("abc.txt", "r")  
print(f.tell())  
f.seek(17)  
print(f.tell())  
x=f.read()  
print(x)
```

```
print(f.tell())
```

```
f.close()
```

Example5:

#Program to demonstrate writelines() method

```
f=open("abc.txt", "w")
```

```
s=["Welcome", "To", "Python"]
```

```
f.writelines(s)
```

```
f.close()
```

Example6:

#Program to demonstrate readlines() method

```
f=open("sample.py", "r")
```

```
lines=f.readlines()
```

```
for line in lines:
```

```
    print(line)
```

```
f.close()
```

By

Mr. Venkatesh Mansani

Naresh i Technologies